

Introduction

Algorithmic trading, or "algo trading," refers to the use of computer programs to execute trades based on pre-defined, mathematical rules. Early algorithmic trading systems relied heavily on rule-based strategies such as moving averages, Bollinger Bands, and Relative Strength Index (RSI) (Chan, 2019). While these methods are easy to interpret, they are rigid and struggle in dynamic market conditions.

The integration of Artificial Intelligence (AI) and Machine Learning (ML) has transformed this field, moving beyond rigid, rule-based execution to creating adaptive, self-learning strategies capable of real-time analysis and decision-making (CFI Trading, 2024). These systems analyse historical and real-time market data and execute buy or sell orders without human intervention. The main objective is to improve trading efficiency, reduce emotional decision-making, and exploit patterns that are difficult for humans to detect.

Algorithmic trading is selected as the problem domain because it is a real-world, data-driven problem that naturally fits AI techniques such as machine learning, neural networks, and reinforcement learning. This literature review explores **how AI techniques are applied across the critical functions of algorithmic trading**, including prediction, strategy execution, and risk management.

1. AI for Market **Prediction and Forecasting**

The primary application of AI in trading is forecasting future market movements. Unlike traditional statistical models, ML algorithms can analyse vast, complex, and noisy datasets to identify subtle, non-linear patterns that are invisible to human traders (Samvakti Journals, 2025).

Prediction Techniques: Deep learning models, particularly Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, are highly effective for time-series forecasting. They capture temporal dependencies in historical price and volume data, allowing them to predict directional movements or future prices with improved accuracy compared to conventional methods (IEEE Xplore, 2024).

Data Integration: Modern predictive models utilise diverse data streams, moving beyond just price and volume. **Natural Language Processing (NLP) and sentiment analysis are deployed to extract market sentiment from news articles, social media, and earnings reports, integrating this unstructured data into the decision-making process to enhance prediction accuracy** (Powers, 2018; Ena Vc, 2024).

2. AI for **Strategy Execution and Optimisation**

AI enhances the speed, efficiency, and adaptability of trade execution, particularly in High-Frequency Trading (HFT). HFT systems execute a large volume of trades in milliseconds to exploit tiny price discrepancies.

High-Frequency Trading (HFT): AI algorithms excel in HFT due to their ability to process microstructure data (individual orders, executions, bid-ask spreads) at ultra-low latency, **making split-second trading decisions that human capacity cannot match** (Ena Vc, 2024; IBKR Quant, 2025).

Optimal Execution: Reinforcement Learning (RL) is increasingly used to optimise trade placement. RL agents learn the optimal sequence of actions (when to buy, sell, or hold) by maximising a cumulative reward function, which often translates to minimising trading costs or market impact over a period. This allows the system to continuously adapt its strategy as market conditions change (CIS UPenn, 2013; Built In, 2024).

3. AI for Risk Management and Stability AI is vital for proactive risk identification and mitigation, which is essential given the speed of algorithmic trading.

Volatility and Risk Assessment: ML models analyse historical market trends and volatility to predict potential adverse price movements, allowing the system to automatically adjust position sizing or place stop-loss orders to protect capital (Ena Vc, 2024).

Anomaly Detection: AI-driven systems are deployed to continuously monitor trading patterns for irregularities. **This helps detect potential market manipulation or technical malfunctions**, contributing to overall market integrity (Jason, 2025).

Conclusion and Challenges While AI transforms algorithmic trading by boosting *efficiency and profitability*, it also introduces systemic challenges. Events like the 2010 Flash Crash highlighted how the rapid, interconnected reactions of numerous algorithms can amplify volatility and create cascading failures (LSE, 2025). Furthermore, the complex, opaque nature of deep learning models leads to a 'black box' problem, necessitating the development of *Explainable AI (XAI)* to build trust and ensure regulatory compliance (Ena Vc, 2024). Ethical concerns such as market manipulation, unequal access to technology, and systemic risk are also important considerations. The future of this domain is centred on leveraging advanced models while upholding ethical standards and transparency.

References

CFI Trading (2024) Algo trading vs AI trading: What are the core differences?

Available at:

<https://cfi.trade/en/uae/educational-articles/trading-essentials/what-are-the-core-differences-between-ai-trading-and-algorithmic-trading> (Accessed: 16 December 2025).

Chan, E. (2019) Algorithmic Trading: Winning Strategies and Their Rationale. Hoboken, NJ: Wiley. Available at:

https://www.google.co.uk/books/edition/Algorithmic_Trading/ClwCTVqEj4oC?hl=en&gbpv=1&dq=Chan (Accessed: 18 December 2025).

Ena Vc (2024) AI in Finance: The Rise of Algorithmic Trading. - Ena Vc. Available at: <https://ena.vc/ai-in-finance-the-rise-of-algorithmic-trading/> (Accessed: 16 December 2025).

IEEE Xplore (2024) Algorithmic Trading: Financial Markets Using Artificial Intelligence. Available at: <https://ieeexplore.ieee.org/document/10593357/> (Accessed: 16 December 2025).

Jason (2025) AI Models in High-Frequency Trading, IBKR Campus US. Available at: <https://www.interactivebrokers.com/campus/ibkr-quant-news/ai-models-in-high-frequency-trading/> (Accessed: 2 January 2026).

LSE (2025) AI and the stock market: are algorithmic trades creating new risks? Available at: <https://www.lse.ac.uk/research/research-for-the-world/ai-and-tech/ai-and-stock-market> (Accessed: 16 December 2025).

Powers, J. (2018) How AI trading technology is making stock market investors smarter — and richer, Built In. Available at: <https://builtin.com/artificial-intelligence/ai-trading-stock-market-tech>. (Accessed: 30 December 2025).

Samvakti Journals (2025) Impact Of AI In Algorithmic Trading Strategies A Comparative Analysis. Available at: https://www.samvaktijournals.com/sjrit.2025.10/impact_ai_algorithmic_trading_strategies_comparative_analysis.pdf (Accessed: 16 December 2025).

Part B – Comparison and Evaluation of AI Techniques

Stated Goal

To develop and implement a supervised machine learning model that classifies the next-day price movement for a specified liquid stock (e.g., NASDAQ: PYPL) as either UP or DOWN based on a sequence of the preceding 60 days of historical prices.

This is a binary time-series classification problem where the model must learn temporal patterns predictive of short-term directional movement.

B1. AI Technique 1: Linear Regression Description : Linear Regression is a supervised learning technique that models the relationship between an independent variable and a dependent variable by fitting a linear equation to observed data. In the context of algorithmic trading, linear regression has traditionally been used to predict future prices or returns based on historical prices or technical indicators (Tsay, 2010).

Application to the Stated Goal

For this problem, linear regression could be adapted to:

- (a) Predict the next-day price numerically
- (b) Convert the predicted value into a class label: UP if predicted price > current price, DOWN otherwise

Strengths: Simple, fast, and highly interpretable. Minimal risk of overfitting with regularisation (e.g., Ridge). Requires little data or computation. Useful as a baseline model

Weaknesses: Assumes linear relationships; cannot capture volatility clustering and momentum shifts. Poor performance on complex, non-linear financial data. Treats observations as independent and ignores the sequential structure. Poor performance on chaotic, noisy financial data (Chen et al., 2023).

Suitability While linear regression is useful as a benchmark, it is not well-suited to capturing the complex, non-linear, and sequential nature of stock price movements.

B2. AI Technique 2: Reinforcement Learning

Reinforcement Learning (RL) involves an agent that learns optimal actions by interacting with an environment and receiving rewards. In trading, RL agents learn trading policies rather than direct predictions (Sutton and Barto, 2018). RL trains an agent to take actions (BUY, SELL, HOLD) to maximise cumulative profit. The reward signal is based on portfolio returns.

Application to the Stated Goal An RL agent could:

- (a) Observe market states
- (b) Execute trading actions (buy/sell/hold)
- (c) Learn policies based on profit and loss

Strengths: Optimises for profit, not just prediction accuracy. Naturally handles transaction costs, risk, and position sizing. Suitable for end-to-end decision-making/trading strategies (Deng et al., 2016). Adaptive to dynamic environments.

Weaknesses: Not a supervised classifier. Does not directly solve the stated goal of classifying UP/DOWN. Requires simulation environment and complex reward engineering. Requires large amounts of data; unstable training/learning in a noisy market. Complex to implement and tune.

Suitability for stated goal: Moderate. While RL is powerful for trading strategy optimisation, it is less suitable for a clearly defined supervised classification task. Example Use: Deng et al. (2016) used deep RL to trade Chinese stocks with risk-adjusted returns outperforming benchmarks, but this goes beyond binary classification.

B3. AI Technique 3: Long Short-Term Memory (LSTM)

LSTM is a type of Recurrent Neural Network (RNN) designed to model sequential data by maintaining a memory of past inputs over long periods. LSTM networks overcome the vanishing gradient problem found in traditional RNNs through the use of memory cells and gating mechanisms to retain relevant information over long sequences (Hochreiter and Schmidhuber, 1997).

Application to the Stated Goal For this project: It processes the 60-day window of OHLCV (Open, High, Low, Close, Volume) data as a time series and outputs a probability of upward movement.

Output: binary classification (UP or DOWN)

Learning task: supervised learning

Strengths: Designed specifically for time-series data. Captures long-term dependencies. Handles non-linear relationships. State-of-the-art for financial time-series forecasting (Fischer and Krauss, 2018).

Weaknesses: Computationally expensive. Requires careful preprocessing and large datasets to avoid overfitting. Requires careful hyperparameter tuning. Less interpretable than linear models.

Suitability for stated goal: Highly suitable. LSTM networks are well-matched to the requirement of learning from rolling 60-day price sequences and performing next-day movement classification.

Example Use: LSTMs achieved 55–60% directional accuracy on S&P 500 constituents using 30–60 day windows (Fischer and Krauss, 2018).

Technique Selection Long Short-Term Memory (LSTM) is selected for in-depth exploration and implementation because: The task involves sequential time-series data. Temporal dependencies are critical (the patterns that make stock prices dependent on a long history). LSTM handles non-linear financial data. The goal is supervised classification. LSTM is covered conceptually in the module

Reference

Chen, W. et al. (2023) 'Deep Learning for Financial Time Series Prediction: A State-of-the-Art Review of Standalone and Hybrid Models', *Computer Modeling in Engineering & Sciences*, 139(1), pp. 187–224. Available at: <https://doi.org/10.32604/cmes.2023.031388>. (Accessed: 18 December 2025)

Deng, Y., Bao, F., Kong, Y., Ren, Z. and Dai, Q., 2016. Deep direct reinforcement learning for financial signal representation and trading. *IEEE Transactions on Neural Networks and Learning Systems*, 28(3), pp.653–664. Available at: <https://doi.org/10.1109/TNNLS.2016.2522401> (Accessed: 28 December 2025)

Fischer, T. and Krauss, C., 2018. Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), pp.654–669. Available at: <https://doi.org/10.1016/j.ejor.2017.11.054> (Accessed: 28 December 2025)

Hochreiter, S. and Schmidhuber, J. (1997) 'Long Short-Term Memory', Neural Computation, 9(8), pp. 1735–1780. Available at: <https://www.bioinf.jku.at/publications/older/2604.pdf> (Accessed: 2 January 2026).

Sutton, R.S. and Barto, A.G. (2018) Reinforcement Learning: An Introduction. 2nd edn. Cambridge, MA: MIT Press. Available at: https://mitpress.ubli.sh.com/book/reinforcement-learning-an-introduction-2?utm_source (Accessed: 2 January 2026).

Tsay, R.S. (2010) Analysis of Financial Time Series. 3rd edn. Hoboken, NJ: John Wiley & Sons.

Data Description: The input will consist of daily "Closing Prices" of PYPL stock, which represents the final trading price of the day.

Data Source: Real-world historical PYPL Data from Yahoo Finance API

Preprocessing Requirements: Min-Max Scaling → 60-Day Sequences (Raw stock prices are highly volatile and must be scaled to a range of (0, 1) using a MinMaxScaler to ensure the Neural Network converges effectively.)

Data Formatting: The data is transformed into a "Sliding Window" format where the past 60 days of prices are used as a feature set to predict the price of the next day.

Dataset Splitting: Dividing the data into Training (80%, to teach the model) and Testing (20%, to evaluate performance) sets.

Get stock data (yfinance)

Process data (pandas, numpy)

Scale data (sklearn)

Build AI model (tensorflow)

Visualize results (matplotlib, seaborn)